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Vehicle control device and vehicle control method

Abstract

A vehicle control device is used for a vehicle that simultaneously transports a user and a package. The vehicle control device includes a controller configured to: acquire an attribute of the user and an attribute of the package; determine whether the user and the package are loaded in the vehicle; and upon determining that the user and the package are loaded in the vehicle, perform air conditioning control according to the attribute of the user and the attribute of the package.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a vehicle control device and a vehicle control method.
BACKGROUND ART
(2) Conventionally, there has been known an invention for controlling an on-board device according to the type of a user's package (Patent Literature 1). The invention described in Patent

Literature 1 controls an air conditioner to lower the temperature in the interior of the vehicle when the package is perceived to be vulnerable to high temperature.

CITATION LIST

Patent Literature

(3) Patent Literature 1: Japanese Patent Application Publication No. 2020-157943

SUMMARY OF INVENTION

Problems to be Solved by Invention

(4) However, the invention described in Patent Literature 1 does not consider the attributes of both the user and the package, although it is necessary to perform air conditioning control in consideration of the attributes of both the user and the package in the case of mixed transportation of passenger and freight in which the user and the package are transported and moved together.

(5) The present invention has been made in view of the above-described problem, and an object the invention is to provide a vehicle control device and a vehicle control method capable of performing air conditioning control in consideration of attributes of both the user and the package.

Solution to Solve Problems

(6) A vehicle control device in accordance with an embodiment of the present invention acquires an attribute of a user and an attribute of a package, determines whether the user and the package are loaded in a vehicle, and performs air conditioning control according to the attribute of the user and the attribute of the package upon determining that the user and the package are loaded in the vehicle.

Effects of Invention

(7) According to the present invention, it is possible to perform air conditioning control in consideration of the attributes of both the user and the package.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic diagram of a system according to an embodiment of the present invention.

(2) FIG. **2** is a functional block diagram of a management server, a communication device, a communication device, and a vehicle according to an embodiment of the present invention.

(3) FIG. **3** is a diagram for explaining attributes of a user and attributes of a package.

(4) FIG. **4** is a sequence chart illustrating an operation example of a system according to an embodiment of the present invention.

(5) FIG. **5** is a sequence chart illustrating an operation example of a system according to an embodiment of the present invention.

(6) FIG. **6** is a flowchart illustrating an operation example of a vehicle control device according to an embodiment of the present invention.

(7) FIG. **7** is a diagram illustrating an example of an air conditioning control method.

(8) FIG. **8** is a diagram illustrating an example of an air conditioning control method.

DESCRIPTION OF EMBODIMENTS

(9) Embodiments of the present invention will now be described with reference to the drawings. In the description of the drawings, the same reference signs are used for the same parts, and the description thereof is omitted.

(10) A configuration example of the system **10** according to the embodiment will be described with reference to FIGS. **1** and **2**. As illustrated in FIG. **1**, the system **10** includes a management server **20**, a communication network **30**, vehicle **40** to **42**, a user **70**, a communication device **60** owned by the user **70**, a package operator **71**, a communication device **61** owned by the package operator **71**, and a package **80** handled by the package operator **71**. FIG. **1** shows three vehicles, but the number of vehicle is not limited thereto. The system **10** may include four or more vehicles.

(11) In this embodiment, the user **70** and the package **80** are loaded in the same vehicle. Such a form is referred to as "integration of passenger and freight transport". More specifically, "integration of passenger and freight transport" is defined as "a form of a transportation of freight and passengers, and an operation together". "Freight" is defined as "goods carried by a transportation means". "Transportation means" includes railways, buses, taxis, aircraft, ships, and the like. In this embodiment, "transportation means" is described as a vehicle. "Goods" refers mainly to package and parcel, the size of which is limited to those which can be loaded in a vehicle. "Goods" include animals (pets), plants, food and general merchandise. "Food" includes food that requires refrigeration (refrigerated food) and food that requires freezing (frozen food). In the "integration of passenger and freight transport", a space for passengers to stay and a space for freight to be loaded are usually clearly separated. In this embodiment, the transportation means is a vehicle, and a seat where passengers sit and a seat where cargo is loaded are clearly separated. Hereinafter, "goods" are referred to as package, and "passengers" are referred to as users. "Package operators" include pet shop employees, flower shop employees, lunch box shop employees, and internet supermarket employees. An internet supermarket is a service that accepts orders via the internet and delivers items directly to homes. In English-speaking countries, an internet supermarket is called an online supermarket or an online grocery. The "packages" in this embodiment includes pets, flowers, lunch boxes, and food items (including refrigerated food).
(12) The management server **20** communicates with the vehicle **40** to **42** and the communication device **60** to **61** via the communication network **30**. The management server **20** is a general-purpose computer (controller) including a central processing unit (CPU) **21**, a memory **22**, a communication I/F **23**, and a storage device **24**, and these components are electrically connected one another via a bus or the like (not illustrated). The management server **20** is used for the dispatch service of the vehicle **40** to **42**. The location of the management server **20** is not particularly limited, but for example, the management server **20** is installed in a management center of an operator operating the vehicle **40** to **42**.
(13) The CPU **21** reads various programs stored in the storage device **24** and the like into the memory **22** and executes various commands contained in the programs. The memory **22** is a storage medium such as Read Only Memory (ROM) and Random Access Memory (RAM). The storage device **24** is a storage medium such as Solid State Drive (SSD) and Hard Disk Drive (HDD). A part (or all) of the system **10** including the functions of the management server **20** described below may be provided by an application (such as Software as a Service (SaaS)) arranged on the communication network **30**.
(14) The communication I/F **23** is implemented as hardware such as a network adapter, various communication software, and a combination thereof, and can realize wired or wireless communication via a communication network **30** and the like. The communication I/F **23** functions as an input unit and an output unit for transmitting and receiving data.
(15) The communication network **30** is described as the Internet without limiting to this, and other wireless communication systems may be employed. The management server **20**, the vehicle **40** to **42**, and the communication device **60**, **61** are connected to the communication network **30** by the Internet.
(16) An example of the vehicle **40** to **42** is a taxi. The vehicle **40** to **42** may be an ordinary vehicle with a driver present or an autonomous vehicle without a driver present. An autonomous vehicle without a driver present may be described as a robotic taxi or an unmanned taxi. In this embodiment, the vehicle **40** to **42** is described as an autonomous vehicle without a driver. As described above, the vehicle **40** to **42** is a vehicle for "integration of passenger and freight transport".
(17) The user **70** requests a vehicle using the communication device **60**. Similarly, the package operator **71** requests a vehicle using the communication device **61**. A vehicle dispatch application (hereinafter simply referred to as a vehicle dispatch app) used for reserving a vehicle is installed in

the communication device **60**, **61**, and the user **70** and the package operator **71** request a vehicle using the vehicle dispatch app.

(18) Next, a detailed configuration of the management server **20**, the vehicle **40**, and the communication device **60** and **61** will be described with reference to FIG. **2**. Although the vehicle **41**, **42** is omitted in FIG. **2**, the vehicle **41** and **42** has the same configuration as the vehicle **40**.

(19) As illustrated in FIG. **2**, the communication device **60** includes a communication I/F **60***a*, a vehicle dispatch app **60***b*, and a GPS receiver **60***c*. Similarly, the communication device **61** includes a communication I/F **61***a*, a vehicle dispatch app **61***b*, and a GPS receiver **61***c*. The communication I/F **60***a* and the communication I/F **61***a* have the same structure as the communication I/F **23**, and communicate with the management server **20** via the communication network **30**. As an example, the communication device **60**, **61** is a portable terminal such as a smartphone or a tablet. The communication device **60**, **61** may be a wearable device. Although not illustrated, the communication device **60**, **61** includes a CPU, a memory, a storage device, and the like in the same manner as the management server **20**.

(20) The vehicle dispatch app **60***b* is used for vehicle requests as described above. The vehicle dispatch app **60***b* functions as a user interface when the user **70** requests a vehicle. The vehicle dispatch app **60***b* is realized by the CPU in the communication device **60** reading and executing a dedicated application program from a storage device in the communication device **60**. When the user **70** uses the vehicle dispatch app **60***b*, that is, when requesting a vehicle, the user enters his/her own information into the vehicle dispatch app **60***b* in advance and registers it. Such pre-registration is a well-known technology and has been adopted in many applications. By performing pre-registration, unique identification information (also referred to as user ID, account ID, or the like.) is given to the user **70**, and the user **70** can request a vehicle using the vehicle dispatch app **60***b*. The information to be entered in advance is, for example, the name, nickname, gender, address, telephone number, email address, payment method and attributes of the user **70**.

(21) The "attributes" to be registered in advance will now be described with reference to FIG. **3**. The "attributes of the user **70**" in this embodiment are information on the allergy of the user **70** and information on the preference on the air in an interior of a vehicle. There are many types of allergies. Examples include allergies caused by pets (such as mites, mold) and allergies caused by plants (such as pollen). The user **70** registers information on allergies as attributes in advance. "Preference on the air in an interior of a vehicle" in this embodiment means the preference of the user **70** on the odor and temperature in an interior of the vehicle. If the user does not like something that emits an odor, this fact is registered as "preference information" in advance. As for the room temperature of the vehicle, the room temperature the user feels comfortable varies from user to user. The user **70** registers the temperature the user feels comfortable as "preference information" in advance. It is noted that the "temperature the user feels comfortable" may vary depending on the season. Therefore, the system may require the user **70** to enter a temperature for each season.

(22) The package operator **71** also registers the attributes of the package **80** to be handled in advance. The "attributes of the package **80**" are information on an allergy of the package **80**, information on an odor of the package **80**, and information on a temperature of the package **80**. When the package **80** is a pet, plant, and the like, the package operator **71** registers that the package **80** has an allergy such as mite, mold, pollen, and the like. If the package **80** is a lunch box emitting an odor, the package operator **71** registers that the package **80** emits an odor. If the package **80** is refrigerated food, the package operator **71** registers that temperature needs to be controlled to 10 degrees or less.

(23) In this embodiment, the air conditioning inside the vehicle is controlled according to "attributes of the user **70**" and "attributes of the package **80**." As illustrated in FIG. **3**, when allergy information is registered as an attribute of the user **70** and allergy information is registered as an attribute of the package **80**, the air conditioning control is performed. When odor information is registered as an attribute of the user **70** and odor information is registered as an attribute of the

package **80**, the air conditioning control is performed. When temperature information is registered as an attribute of the user **70** and temperature information is registered as an attribute of the package **80**, the air conditioning control is performed. As illustrated in FIG. **3**, the air conditioning control is not performed in any other combination. There are users who do not have allergies, users who do not care about odors, and users who accept any temperature. Such users do not have to register attributes. For a user who has not registered the attribute ("no attribute" in FIG. **3**), air conditioning control is not performed regardless of the attribute of the package. There are packages that do not cause allergy, packages that do not emit odor, and packages that do not require temperature control. The package operator **71** does not have to register the attributes of such packages. For packages whose attributes are not registered ("no attributes" in FIG. **3**), air conditioning control is not performed regardless of the attributes of the user. However, what is not performed is air conditioning control according to the attributes. Normal air conditioning control is performed. "Normal air conditioning control" refers to air conditioning control set in advance according to the outside temperature and the like.

(24) After the pre-registration is completed, the user **70** enters a desired place to be loaded into the vehicle, desired time to be loaded into a vehicle, desired destination (place to be unloaded from a vehicle), desired arrival time, desired seat, and the like into the vehicle dispatch app **60***b*, and requests a vehicle. For example, the user **70** selects a desired seat from among available seats (unreserved seats). Selection of a seat is optional. When the user **70** does not select a seat, a seat is automatically selected by the management server **20**. The vehicle dispatch app **60***b* transmits a vehicle dispatch request to the management server **20** according to an input of the user **70**. Further, the communication device **60** displays various information (such as vehicle dispatch request receipt, estimated arrival time, estimated route) included in the signal returned from the management server **20** in response to the vehicle dispatch request on a display. However, the method of implementing the vehicle dispatch app **60***b* is not limited thereto. For example, the communication device **60** may access the server that provides the function of the vehicle dispatch app **60***b*, receive the function provision, and display the execution result of the function transmitted from the server by a browser. Similarly, after the pre-registration is completed, the package operator **71** enters the desired place to be loaded into the vehicle, desired time to be loaded into the vehicle, desired destination (place to be unloaded), desired arrival time, desired seat, and the like into the dispatch app **61***b*, and requests a vehicle.

(25) The position information of the communication device **60** acquired by the GPS receiver **60***c* is transmitted to the management server **20** at any timing. Similarly, the position information of the communication device **61** acquired by the GPS receiver **61***c* is transmitted to the management server **20** at any timing.

(26) As illustrated in FIG. **2**, the vehicle **40** includes a communication I/F **401**, a vehicle electronic control unit (ECU) **402**, a camera **403**, a seating sensor **404**, a heating, ventilation, and air conditioning (HVAC) system **405**, and a GPS receiver **406**. The communication I/F **401** has the same configuration as the communication I/F **23** and communicates with the management server **20** via the communication network **30**. The vehicle ECU **402** is a computer for controlling the vehicle **40**. The vehicle ECU **402** controls various actuators (such as brake actuators, accelerator actuators, steering actuators) based on commands received from the management server **20**. The camera **403** is installed in the interior of the vehicle **40**. The camera **403** detects the seating position of the user **70** (the position of the seat on which the user **70** is seated). Further, the camera **403** detects the position of the seat on which the package **80** is placed. These positions may be detected by the seating sensor **404** instead of the camera **403**. The HVAC system **405** controls the temperature, humidity, and airflow in the interior of the vehicle **40** and includes a cooling device, a heating device, a blower, and the like. The position information of the vehicle **40** acquired by the GPS receiver **406** is transmitted to the management server **20** at any timing.

(27) As illustrated in FIG. **2**, the CPU **21** of the management server **20** includes, as an example of a

plurality of functions, a vehicle dispatch reception **211**, an allocator **212**, a travel route calculator **213**, and an air conditioning control method generator **214**. A map database **241** and a customer database **242** are stored in a storage device **24** of the management server **20**.
(28) The map database **241** stores map information necessary for a route guidance such as road information and facility information. The map information includes the number of lanes of a road, road width information, and road undulation information. Further, the map information includes road signs indicating speed limits, one-way traffic, crosswalks, and markings. The map information stored in the map database **241** may be high-precision map data (HD MAP) or normal map data (SD MAP).
(29) The customer database **242** stores information (such as name, attributes) of the user **70** and the package operator **71**, a vehicle usage history, and the like.
(30) The vehicle dispatch reception **211** receives a vehicle dispatch request of the user **70** entered into the communication device **60** (vehicle dispatch app **60***b*). The vehicle dispatch reception **211** receives a vehicle dispatch request entered into the communication device **61** (vehicle dispatch app **61***b*) by the package operator **71**. The vehicle dispatch reception **211** has a function of notifying the communication device **60** that the vehicle dispatch request of the user **70** has been received, the scheduled arrival time at the place to be loaded into the vehicle, the scheduled travel route to the destination, and the like. Similarly, the vehicle dispatch reception **211** has a function of notifying the communication device **61** that the vehicle dispatch request of the package operator **71** has been received, the scheduled arrival time at the place to be loaded into the vehicle, the scheduled travel route to the destination, and the like.
(31) The vehicle dispatch reception **211** acquires position information of the user **70** from the communication device **60**, acquires position information of the package operator **71** from the communication device **61**, and acquires the position information of the vehicle **40** from the vehicle **40**. "Position information of the user **70**" means the position information of the communication device **60** possessed by the user **70**. "Position information of the package operator **71**" means the position information of the communication device **61** possessed by the package operator **71**. The vehicle dispatch reception **211** outputs the acquired position information to the allocator **212** and the travel route calculator **213**.
(32) The allocator **212** allocates an appropriate vehicle from a plurality of vehicles **40** to **42** based on the received vehicle dispatch request. For example, the allocator **212** can allocate the vehicle closest to the desired place to be loaded into the vehicle for the user **70** or the package operator **71** from the plurality of vehicles **40** to **42** to improve efficiency. In this embodiment, a description is given, supposing that the vehicle **40** is allocated.
(33) The travel route calculator **213** calculates the travel route from the current location of the vehicle **40** to the destination via the place to be loaded into the vehicle by referring to the location information obtained from the vehicle dispatch reception **211** and the map database **241**. In this embodiment, it is assumed that the user **70** and the package **80** share a ride in order to improve the operation efficiency of the vehicle **40**. Regarding the ride-sharing, it is possible to confirm with the user **70** and the package operator **71** whether to permit the ride-sharing, but the ride-sharing will be described as being permitted. It does not matter whether the user **70** is in the vehicle first or the package **80** is loaded the vehicle first. If the destinations of the user **70** and the package **80** are in the same direction, the efficient operation of the vehicle **40** can be realized, but for the application of the present invention, the destinations need not be in the same direction.
(34) The air conditioning control method generator **214** acquires the attributes of the user **70** and the package **80** by referring to the customer database **242**. The air conditioning control method generator **214** generates the air conditioning control method according to the attributes of the user **70** and the package **80**. The air conditioning control method generator **214** transmits the generated air conditioning control method to the HVAC system **405**.
(35) Next, an operation example of the system **10** will be described with reference to a sequence

charts of FIGS. **4** and **5**. In stop S**101**, the user **70** requests a vehicle using the communication device **60**. A request signal is transmitted from the communication device **60** to the management server **20**. In step S**103**, the package operator **71** requests a vehicle using the communication device **61**. A request signal is transmitted from the communication device **61** to the management server **20**. In step S**105**, the management server **20** allocates a vehicle and calculates a travel route based on the received request. The process continues to step S**107** and the management server **20** acquires the attributes of the user **70** and the package **80** by referring to the customer database **242**. The process continues to step S**109** and the management server **20** generates an air conditioning control method according to the attributes of the user **70** and the package **80**. The management server **20** transmits a vehicle dispatch plan including a travel route and the like, and the air conditioning control method to the vehicle **40**.

(36) In step S**111**, the vehicle **40** acquires the vehicle dispatch plan and the air conditioning control method from the management server **20**. The process continues to step S**113** and the vehicle ECU **402** controls various actuators to allow the vehicle **40** travel along the travel path. The vehicle **40** heads to the desired place to be loaded for the user **70** to load the user **70** into the vehicle. Further, the vehicle **40** heads to a desired loading place of the package **80** and loads the package **80**. The process continues to step S**115** and the camera **403** or the seat sensor **404** detects the position of the seat on which the user **70** is seated and the position of the seat on which the package **80** is placed. The process continues to step S**117** and the HVAC system **405** controls the air conditioning using the air conditioning control method acquired from the management server **20**. The process continues to step S**119** and the package is unloaded at a desired unloading place of the load **80**, and the air conditioning control is completed. Details of the processes in steps S**107**, **109**, **115**, and **117** will be described with reference to FIGS. **6** to **8**.

(37) First, details of step S**107** will be described. In step S**201** in FIG. **6**, when the management server **20** acquires attributes of the user **70** and the package **80**, it first acquires attributes of the package **80**. That is, the package **80** comes first and the user **70** comes later in the acquisition order of attributes. The management server **20** determines whether the attributes of the package **80** may affect the user **70**. As described in FIG. **3**, if the attributes of the package **80** include any of an allergy, an odor, and a temperature, the management server **20** determines that the attributes of the package **80** may affect the user **70** (“YES” in step S**203**). If it is determined to be “NO” in step S**203**, that is, there is no attribute of the package **80** (the attributes of the package **80** is not registered), a normal air conditioning control is performed (step S**205**).

(38) In step S**207**, the management server **20** acquires the attributes of the user **70**. The process continues to step S**209** and the management server **20** determines whether the attributes of the user **70** matches the attributes of the package **80**. The “attribute match” occurs when a pollen allergy is registered as an attribute of the user **70** and a plant dispersing pollen is registered as an attribute of the package **80**. Alternatively, the “attribute match” occurs when an odor aversion is registered as an attribute of the user **70** and a lunch box that emits an odor is registered as an attribute of the package **80**. Alternatively, the “attribute match” occurs when an interior temperature is registered as an attribute of the user **70** and a food that needs refrigeration is registered as an attribute of the package **80**. Alternatively, the “attribute match” may be defined as a case in which keywords related to attributes of the user **70** and the package **80** match. Alternatively, the “attribute match” may be defined as a case in which properties, features, characteristics, and the like related to attributes of the user **70** and the package **80** match. If the attributes of the user **70** match those of the package **80** (YES in step S**209**), the process continues to step S**211** and the management server **20** performs air conditioning control depending to the attributes of the user **70** and the package **80**. The determination of matching is performed in the process in step S**107** (FIG. **4**).

(39) Next, the details of step S**115** will be described. FIG. **7** shows the interior of the vehicle **40**. Reference signs **90** and **91** denote an air conditioning device with an HVAC system **405** including a cooling device, a heating device, a blower, and the like. Reference sign **92** denotes an exhaust port.

Reference signs **93** to **96** denote a seat. As illustrated in FIG. **7**, it is assumed that the user **70** is seated on a seat **96** and the package **80** is placed on a seat **94**. In this case, the camera **403** or the seating sensor **404** detects the position of the seat on which the user **70** is seated and the position of the seat on which the package **80** is placed.

(40) Next, the details of steps S**109** and S**117** will be described. The air conditioning control method generator **214** generates an air conditioning control method based on the position of the seat detected by the camera **403** or the seating sensor **404**. In FIG. **7**, it is assumed that a pollen allergy is registered as an attribute of the user **70** and a plant dispersing pollen is registered as an attribute of the package **80**. In this case, the attribute of the user **70** and the attribute of the package **80** match. The air conditioning control method generator **214** generates an air conditioning control method so that the influence of the package **80** on the user **70** is reduced. A method of blowing air from the blower **90** and **91** toward the exhaust port **92** as illustrated in FIG. **7** is an example of the air conditioning control method. With this method, an influence of the allergen of the package **80** (plant) on the user **70** can be reduced. In another example, it is assumed that the user **70** has a registered attribute of an odor aversion, and a lunch box that emits odor has been registered as an attribute of the package **80**. In this case, a method of blowing air from the blower **90** and **91** toward the exhaust port **92** as illustrated in FIG. **7** is also applied. Thus, the influence of the odor of the package **80** (lunch box) on the user **70** can be reduced. The air may be blown from the blower **90** and **91** toward the user **70**. Thus, the air around the user **70** always flows, and the influence of an allergen or an odor can be reduced. Furthermore, the air may be blown so that the user **70** is located upstream and the package **80** is located downstream relative to the airflow. It should be noted that reducing an influence on the user **70** includes a case in which there is no influence on the user **70**.

(41) As another example, it is assumed that an interior temperature is registered as an attribute of the user **70** and a food that needs refrigeration is registered as an attribute of the package **80**. An air curtain illustrated in FIG. **8** is an example of an air conditioning control method in this case. Reference sign **100** in FIG. **8** indicates an air curtain generator (a type of HVAC system **405**), and reference sign **101** indicates an air curtain. An “air curtain” generates a film of fast airflow to block air movement. The term “block” is used not only to completely block air movement but also to suppress air movement. The “air curtain” is provided to block air conditioning and prevent the intrusion of odors, enabling a use of a space without providing walls or partitions. The air curtain **101** in this embodiment generates a film of airflow from a ceiling to a floor. As illustrated in FIG. **8**, the interior of the vehicle **40** is separated into regions **102** and **103** by the air curtain **101**. The region **102** is cooled to 10 degrees or less by a cooling device. Thus, the package **80** (refrigerated food) is kept at an appropriate temperature. Cold air circulating in the region **102** by the air curtain **101** does not flow into the region **103**. Thus, the user **70** can reduce the influence of cold air. Thus, even a user who does not like cold air can be loaded into the vehicle **40** at the same time as refrigerated food. The region **103** can be adjusted to a desired temperature for the user **70**. For example, if the desired temperature for the user **70** is 23 degrees, the HVAC system **405** can be controlled so that the region **103** is at 23 degrees. Even if the area **103** is set to 23 degrees, the area **102** is not influenced by the air curtain **101** and is kept at 10 degrees or less. It is noted that the air curtain **101** can also be applied to the above-mentioned allergies and odors. In place of the air curtain **101**, a physical shield such as a shutter or a partition board may be provided to block air movement.

Operation and Effect

(42) As described above, according to this embodiment, the following operation and effect can be obtained.

(43) The management server **20** (controller) acquires the attributes of the user **70** and the attributes of the package **80**. Acquisition of the attributes is realized by referring to the customer database **242**. The management server **20** determines whether the user **70** and the package **80** are loaded in the vehicle. This determination is made based on data detected by the camera **403** or the seating

sensor **404**. When the management server **20** determines that the user **70** and the package **80** are loaded in the vehicle, it performs air conditioning control according to the attributes of the user **70** and the package **80**. Thus, it is possible to perform air conditioning control considering the attributes of both the user **70** and the package **80**.
(44) The attributes of the user **70** include information on a prescribed allergy or information indicating a preference on the air in the interior of the vehicle. "Information indicating a preference on the air in the interior of the vehicle" is information indicating a preference of the user **70** on the odor and temperature in the interior of the vehicle. Attributes of the package **80** include information on a prescribed allergy, information on an odor, or information indicating that temperature control is required.
(45) Air conditioning control includes at least one of the following: controlling an airflow to reduce the influence of the package **80** on the user **70**; blocking a space between the package **80** and the user **70** with the airflow; and blocking the space between the package **80** and the user **70** with a physical device. The management server **20** selects one of the three air conditioning controls according to the attributes of the user **70** and the package **80** to perform air conditioning control. Thus, the user **70** is not influenced by the package **80**, and a comfortable interior space can be provided to the user **70**.
(46) Each of the functions described in the above embodiments may be implemented by one or more processing circuits. The one or more processing circuits include a programmed processing device such as a processing device including an electric circuit. Further, the one or more processing circuits include a device such as an application-specific integrated circuit (ASIC) or circuit component arranged to perform each of the described functions.
(47) Although embodiments of the present invention have been described above, the statements and drawings that form a part of this disclosure should not be understood as limiting the present invention. The present disclosure will reveal to those skilled in the art a variety of alternative embodiments, embodiments, and operational techniques.
(48) Although the management server **20** is described as a vehicle control device, the vehicle control device is not limited to the management server **20**. The configuration and functions of the management server **20** may be mounted on the vehicle **40**. In this case, the controller mounted on the vehicle **40** functions as a vehicle control device. The attributes of the package **80** need not necessarily be registered in advance. The package operator **71** may enter the attributes of the package **80** when reserving a vehicle. Furthermore, the vehicle **40** may be provided with a sensor for detecting an odor, and the sensor may detect whether the package **80** emits an odor when loaded.
REFERENCE SIGNS LIST
(49) **20** management server **211** vehicle dispatch reception **212** allocator **213** travel route calculator **214** air conditioning control method generator

Claims

1. A vehicle control device used for a vehicle that simultaneously transports a user and a package, the vehicle control device comprising a controller, wherein the controller is configured to: acquire an attribute of the package; determine whether the attribute of the package may affect the user; determine that the attribute of the package may not affect the user in a case where the attribute of the package is not registered; upon determining that the attribute of the package may affect the user, acquire an attribute of the user and determine whether the attribute of the user matches the attribute of the package; determine whether the user and the package are loaded in the vehicle; upon determining that the attribute of the package may not affect the user and upon determining that the user and the package are loaded in the vehicle, perform a normal air conditioning control; and upon determining that the attribute of the user matches the attribute of the package and upon

determining that the user and the package are loaded in the vehicle, perform air conditioning control according to the attribute of the user and the attribute of the package.

2. The vehicle control device according to claim 1, wherein the attribute of the user includes information on a prescribed allergy or information indicating a preference on air in an interior of the vehicle; and wherein the attribute of the package includes information on a prescribed allergy, information on an odor, or information indicating requirement of temperature control.

3. The vehicle control device according to claim 2, wherein the air conditioning control includes at least one of: (a) controlling airflow to reduce an influence of the package on the user; (b) blocking a space between the package and the user with airflow; and (c) blocking the space between the package and the user with a physical device; and the controller is configured to select one of the aforementioned (a) to (c) to perform the air conditioning control according to the attribute of the user and the attribute of the package.

4. A vehicle control method of a vehicle control device used for a vehicle that simultaneously transports a user and a package, the vehicle control method comprising: acquiring an attribute of the package; determining whether the attribute of the package may affect the user; determining that the attribute of the package may not affect the user in a case where the attribute of the package is not registered; upon determining that the attribute of the package may affect the user, acquiring an attribute of the user and determining whether the attribute of the user matches the attribute of the package; determining whether the user and the package are loaded in the vehicle; upon determining that the attribute of the package may not affect the user and upon determining that the user and the package are loaded in the vehicle, performing a normal air conditioning control; and upon determining that the attribute of the user matches the attribute of the package and upon determining that the user and the package are loaded in the vehicle, performing an air conditioning control according to the attribute of the user and the attribute of the package.